Inorganic CUME

February 15, 2019

Answer any five questions below. Only the first five of your answers will be graded.

1. (20 points) During electrochemical deposition of different metals such as Cu, Zn or Ag, the deposited films are never smooth or "shinning" if no levelers or additives are added during the electrochemical process. Why is that? What are the potential solutions to eliminate the dendritic microstructures during electrochemical deposition?

2. (20 points) Explain the roles of capping agents used during synthesis. What are the differences of thermodynamically controlled and dynamically controlled crystal growth? Give an example of each.

3. (20 points) What are a) surface-enhance Raman scattering, b) surface-enhanced fluorescence, c)X-ray photoelectron spectroscopy (XPS) and d) X-ray diffraction (XRD). Briefly explain the working principle of each. What is the Bragg's Law?

4. (20 points) A copper substrate has been covered by a thin film of inorganic nanocrystals. List at least three different characterization techniques that can be combined to study the properties of coating layer including the chemical composition, structure, surface area, thickness and uniformity.

5. (20 points) Explain what is (a) Schottky defect, (b) Frenkel defect and (c) Stacking Fault. What are the differences between intrinsic and extrinsic defects?

6. (20 points) Iron(II) forms a complex in hemoglobin. For each of the two octahedral complex ions $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$, draw an energy diagram showing orbital splitting, predict the number of unpaired electrons, and identify the ion as low spin or high spin.

7. (20 points) For each of the following reactions, use the given rate law to determine the reaction order with respect to each reactant and the overall order.
   (a) $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$; rate = $k[NO]^2[O_2]$
   (b) $CH_3CHO(g) \rightarrow CH_4(g) + CO(g)$; rate = $k[CH_3CHO]^{3/2}$
   (c) $H_2O_2(aq) + 3I^-(aq) + 2H^+(aq) \rightarrow I_3^-(aq) + 2H_2O(l)$; rate = $k[H_2O_2][I^-]$

8. (20 points) For the reaction $N_2O_4(g) = 2NO_2(g)$, $K_c = 0.21$ at 100°C. At a point during the reaction, $[N_2O_4] = 0.12\ M$ and $[NO_2] = 0.55\ M$. Is the reaction at equilibrium? If not, in which direction is it progressing?

9. (20 points) Give a formula (or sketch) for each common organic ligand that can act as a) one-electron donor, b) two-electron donor, c) three-electron donor, and d) four-electron donor.

10. (20 points) Why diamond is electrically insulating and hard, while graphite is electrically conductive and slippery, although they are both made of pure carbon? Which one, diamond or graphite, is the most stable form at standard thermodynamic conditions? What is the difference between quartz and glass in structural chemistry?